Project Planning Phase
Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	10 July 2026
Team ID	SWTID-2026-6119
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

The backlog below lays out each sprint's stories along with their sizing and ownership.

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Data Collection	USN-1	As a data scientist, I want to pull together a historical credit card approval dataset so the model has real data to learn from.	2	High	Naveen
Sprint-1		USN-2	As a data scientist, I want to load that dataset into a Pandas DataFrame so I can start exploring it.	1	High	Naveen
Sprint-1	Data Preprocessing	USN-3	As a data scientist, I want to handle missing values in the dataset so training runs on clean, complete records.	3	High	Manogna
Sprint-1		USN-4	As a data scientist, I want to encode the categorical columns so the model can actually read them as numbers.	3	High	Manogna
Sprint-1		USN-5	As a data scientist, I want to scale the numerical features so every input reaches the model on a consistent range.	3	Medium	Manogna
Sprint-1		USN-6	As a data scientist, I want to strip out duplicate rows and outliers so noisy records don't skew the model.	3	Medium	Manogna
Sprint-1	Exploratory Data Analysis	USN-7	As a data scientist, I want count plots and a correlation heatmap so I can see how the features relate to each other.	3	Medium	Naveen
Sprint-1		USN-8	As a data scientist, I want to split the data into training and testing sets so I can measure performance honestly.	2	High	Naveen
Sprint-2	Model Training	USN-9	As a data scientist, I want to train a Logistic Regression model on the cleaned data as a baseline.	3	High	Manogna
Sprint-2		USN-10	As a data scientist, I want to train a Decision Tree model on the same data to compare against the baseline.	3	High	K Akshaya
Sprint-2		USN-11	As a data scientist, I want to train a Random Forest model to see whether an ensemble improves results.	3	High	K Akshaya

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-2		USN-12	As a data scientist, I want to train an XGBoost model to test a boosted-tree approach as well.	3	High	K Akshaya
Sprint-2	Model Evaluation	USN-13	As a data scientist, I want to score every trained model on Accuracy, Precision, Recall and F1-Score so the comparison is objective.	5	High	K Akshaya
Sprint-2		USN-14	As a data scientist, I want to line up the models against each other and pick the strongest one to deploy.	3	High	K Akshaya
Sprint-3	Flask Application	USN-15	As a developer, I want the Flask project scaffolded with its core routes so there's a foundation to build the app on.	3	High	K Akshaya
Sprint-3		USN-16	As a developer, I want a prediction form built with HTML and Bootstrap so applicants have a clean page to fill in.	3	High	K Akshaya
Sprint-3		USN-17	As a developer, I want the saved model wired into the Flask backend so submitted forms actually produce a prediction.	5	High	K Akshaya
Sprint-3		USN-18	As a developer, I want the prediction shown on its own results page so the outcome is easy to find.	2	High	K Akshaya
Sprint-3	Error Handling	USN-19	As a developer, I want form input validated and errors handled cleanly instead of the app breaking.	3	Medium	K Akshaya
Sprint-3		USN-20	As a developer, I want a custom 404 page so users hitting a bad URL still see something sensible.	2	Low	K Akshaya
Sprint-3	Testing	USN-21	As a QA engineer, I want to run the whole prediction flow end-to-end to confirm it works as intended.	2	High	K Akshaya
Sprint-4	Deployment	USN-22	As a developer, I want the Flask app deployed locally and on IBM Cloud so it's reachable outside my machine.	5	High	K Akshaya
Sprint-4	Documentation	USN-23	As a technical writer, I want the project report and README written up so the work is properly documented.	3	Medium	Team (All)
Sprint-4		USN-24	As a technical writer, I want a project PPT and demo video ready so the work can be presented clearly.	3	Medium	Team (All)
Sprint-4	Testing	USN-25	As a QA engineer, I want to run performance and user-acceptance testing on the live app before we call it done.	5	High	K Akshaya
Sprint-4	Final Submission	USN-26	As a team, we want the GitHub repository finalized and submitted through SmartBridge SkillWallet to close out the project.	4	High	Team (All)

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	16 Jun 2026	21 Jun 2026	20	21 Jun 2026
Sprint-2	20	6 Days	23 Jun 2026	28 Jun 2026	20	28 Jun 2026
Sprint-3	20	6 Days	30 Jun 2026	05 Jul 2026		
Sprint-4	20	6 Days	07 Jul 2026	12 Jul 2026		

Velocity:

For the Credit Card Approval Prediction System, work was planned across four sprints of six days each, adding up to 80 story points over the whole backlog - an average of 20 points per sprint. Velocity therefore works out to Total Story Points Completed divided by Number of Sprints, which is 80 / 4 = 20 story points per sprint, and that figure was used to size the remaining sprints and keep the project on pace for the internship deadline.

$$V = \frac{\text{Total Story Points Completed}}{\text{Number of Sprints}} = \frac{80}{4} = 20 \text{ points/sprint}$$

Burndown Chart:

A burndown chart plots remaining work against time, giving an at-a-glance read on whether a sprint is on pace to finish. It's a staple of agile methods like Scrum, though it applies just as well to any project where progress can be measured incrementally.

https://www.visual-paradigm.com/scrum/scrum-burndown-chart/

https://www.atlassian.com/agile/tutorials/burndown-charts

Reference:

https://www.atlassian.com/agile/project-management

https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software

https://www.atlassian.com/agile/tutorials/epics

https://www.atlassian.com/agile/tutorials/sprints

https://www.atlassian.com/agile/project-management/estimation

https://www.atlassian.com/agile/tutorials/burndown-charts